

Heejun Shin

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INTERESTS	Causal Inference, Bayesian Nonparametric Models, Environmental Statistics	
EMPLOYMENT	Assistant Professor , Konkuk University, Department of Applied Statistics Postdoctoral Research Fellow , Harvard University Mentor: Professor Francesca Dominici Statistical Consultant , Harvard Data Science Initiative “Real-Time AI Tool for Severe Event Prevention” in partnership with GE Vernova	2026–present 2024–2026 2025–2026
EDUCATION	Ph.D. in Statistics, University of Florida Thesis: Causal Inference with Bayesian Modeling for Challenging Environmental Health Problems Advisor: Professor Joseph Antonelli B.S. in Applied Statistics, Konkuk University Ranked first in class	2024 2019
PREPRINTS	Shin, H. , Braun, D., Irene, K., and Antonelli, J. (2023+). A spatial interference approach to account for mobility in air pollution studies with multivariate continuous treatments. arXiv:2305.14194 . <i>Under Revision, Journal of the American Statistical Association</i> [Link] • Winning paper of the 2025 ASA Epidemiology Section Norman Breslow Prize. • Winning paper of the 2025 ASA HPSS Section Student Paper Competition. (Declined)	
PUBLICATIONS	[3] Antonelli, J., Shin, H. , Kang, S., Franks, A., Audirac, M., Braun, D. (2025) Robust Statistical Approaches to Understanding the Causal Effect of Air Pollution Mixtures. <i>Health Effects Institute</i> . [Link] [2] Shin, H. , Linero, A., Audirac, M., Irene, K., Braun, D., and Antonelli, J. (2025). Treatment effect heterogeneity and importance measures for multivariate continuous treatments. <i>The Annals of Applied Statistics</i> , 19(3): 1847-1867. [Journal][arXiv] • Winning paper of the 2024 ASA Biometrics Section Early Career Paper Awards. • Winning paper of the 2024 ASA ENVR Section Student Paper Competition. (Declined) [1] Shin, H. and Antonelli, J. (2023). Improved inference for doubly robust estimators of heterogeneous treatment effects. <i>Biometrics</i> , 79(4): 3140–3152. [Journal][arXiv] • Winning paper of the 2022 ENAR John Van Ryzin Award. * indicates advisee/mentee † indicates corresponding author	
HONORS & AWARDS	IMS New Researcher Travel Award , Institute of Mathematical Statistics The Norman Breslow Prize for the top paper among the winning papers of the Early Career Paper Awards, American Statistical Association Statistics in Epidemiology Section Early Career Paper Award , American Statistical Association Biometrics Section Kenneth and Janet Keene Endowed Dissertation Fellowship , College of Liberal Arts and Sciences, University of Florida Mark CK Yang Student Research Award , UF Joint Statistics and Biostatistics Workshop Boyd Harshburger Travel Award funded by National Science Foundation, The Southern Regional Council on Statistics The John Van Ryzin Award for the top paper among the winning papers of the Distinguished Student Paper Awards, International Biometric Society Eastern North American Region College of Liberal Arts and Sciences Graduate Student Travel Award , University of Florida - Joint Statistical Meetings - Eastern North American Region Spring Meeting, Joint Statistical Meetings	2025 2025 2024 2024 2023 2022 2022 2023 2022

	The President's Award for an outstanding undergraduate student, Korean Statistical Society	2019
TEACHING EXPERIENCE	Instructor , Konkuk University	
	Nonparametrics	2026
	Aanalytical Programming 2	2026
	Guest Lecturer , Harvard University	2024
	BST228 - Applied Bayesian Analysis	
	Instructor , University of Florida	2023
	STA3024 - Introduction to Statistics II	
	Teaching Assistant , University of Florida	2019–2023
	STA3024 - Introduction to Statistics II	STA4322 - Introduction to Statistical Theory
	STA4241 - Statistical Learning	STA4502 - Nonparametric Statistical Methods
	STA4321 - Introduction to Probability	STA6166 - Statistical Methods in Research
MENTORING EXPERIENCE	Harvard University	
	Suhwan Bong, Ph.D. Student, Harvard University	2025–2026
	Salome Kakhaia, visiting Ph.D. student, Utrecht University	2025–2026
INVITED PRESENTATIONS	Seminar talk. “A Spatial Interference Approach to Account for Mobility in Air Pollution Studies with Multivariate Continuous Treatments” <i>London School of Hygiene & Tropical Medicine</i> . Virtual 2025.	
	Seminar talk. “Treatment effect heterogeneity and importance measures for multivariate continuous treatments” <i>Konkuk University</i> . Seoul, Korea 2024.	
	Seminar talk. “Improved inference for doubly robust estimators of heterogeneous treatment effects” <i>WebENAR</i> . Virtual 2024.	
	Seminar talk. “Identifying Interactions between Covariates and Continuous Exposures via Targeted Smoothing” <i>UFSTAT Student Seminar Series</i> . Gainesville, FL. 2023.	
CONTRIBUTED PRESENTATIONS	Contributed talk. “Data-driven Principal Stratification for Continuous Exposure and Mediators” <i>International Biometric Conference</i> . Seoul, Korea 2026.	
	Contributed poster. “Data-driven Principal Stratification for Continuous Exposure and Mediators” <i>The 2026 ISBA World Meeting</i> . Nagoya, Japan. 2026.	
	Contributed talk (as the Norman Breslow Prize winner). “A spatial interference approach to account for mobility in air pollution studies” <i>Joint Statistical Meetings</i> . Nashville, TN. 2025.	
	Contributed talk (as an Early Career Paper Award recipient). “Treatment effect heterogeneity and importance measures for multivariate continuous treatments” <i>Joint Statistical Meetings</i> . Portland, OR. 2024.	
	Contributed talk. “A spatial interference approach to account for mobility in air pollution studies with multivariate continuous treatments” <i>Bayesian Young Statisticians Meeting</i> . Virtual. 2023.	
	Contributed talk. “A spatial interference approach to account for mobility in air pollution studies with multivariate continuous treatments” <i>UF Joint Statistics and Biostatistics Workshop</i> . Gainesville, FL. 2023.	
	Contributed talk. “A spatial interference approach to account for mobility in air pollution studies with multivariate continuous treatments” <i>Joint Statistical Meetings</i> . Toronto, Canada. 2023.	
	Contributed poster. “Causal Effects of Continuous Exposures in the Presence of Spatial Interference: the Effects of Air Pollution on Public Health.” <i>American Causal Inference Conference</i> . Austin, TX. 2023.	
	Contributed poster. “Improved inference for doubly robust estimators of heterogeneous treatment effects.” <i>The Southern Regional Council on Statistics Summer Research Conference</i> . Jekyll Island, GA. 2022.	
	Contributed talk. “Improved inference for doubly robust estimators of heterogeneous treatment effects.” <i>Joint Statistical Meetings</i> . Washington, D.C. 2022.	
	Contributed talk (as the John Van Ryzin Award winner). “High-dimensional and nonparametric Bayesian methodology for treatment effect heterogeneity.” <i>Eastern North American Region Spring Meeting</i> . Houston, TX. 2022.	
SOFTWARE	<i>SepBART</i> : An R package for estimating heterogeneous treatment effects of multivariate continuous exposures (https://github.com/hshin111/SepBART)	

SERVICE

Organizer, *UFSTAT Student Seminar Series*, Department of Statistics, University of Florida 2022-2023
Paper Competition Judge, *ASA Section on Statistics in Epidemiology*
Journal Referee, *Annals of Applied Statistics*, *Biometrics*, *GeoHealth*, *Harvard Data Science Review*, *PeerJ: life and environment*, *Stat*, *Statistics in Biosciences*, *Statistics in Medicine*